

Week 3 Technical Progress Report: Autonomous Agent Environment Deployment

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May 20, 2026

1 Research Objective

The goal for Week 3 was to transition from conceptual taxonomy to a functional autonomous agent environment. The aim was to deploy a local sandbox using `OpenHands` and integrate `Ollama` as the LLM engine to facilitate an autonomous "Think-Act-Observe" loop.

2 Implementation & Commands Executed

To establish the research environment, the following technical steps were executed.

2.1 Sandbox Initialization

The `OpenHands` framework was deployed using a persistent Docker container mapped to a local Ubuntu/WSL filesystem.

Listing 1: Docker Sandbox Deployment

```
export WORKSPACE_BASE=$(pwd)/workspace
docker run -it --rm --pull=always \
  -e SANDBOX_VOLUMES="$WORKSPACE_BASE:/workspace:rw" \
  -v /var/run/docker.sock:/var/run/docker.sock \
  -p 3000:3000 \
  --add-host host.docker.internal:host-gateway \
  ghcr.io/all-hands-ai/openhands:latest
```

2.2 Local Model Integration (Ollama)

Due to network restrictions, the model was fetched via `curl` and registered to the local Ollama engine:

Listing 2: Model Acquisition and Registration

```
# Model Acquisition
curl.exe -L -o llama3-agent.gguf https://huggingface.co/bartowski/
  /Meta-Llama-3.1-8B-Instruct-GGUF/resolve/main/Meta-Llama-3.1-8
  B-Instruct-Q4_K_M.gguf
```

```
# Model Manifest Registration
echo FROM "D:/Ollama/llama3-agent.gguf" > Modelfile-llama
ollama create llama3-agent -f Modelfile-llama
```

3 Experimental Failures & Error Analysis

The deployment revealed three distinct failure points that prevent stable autonomous agent execution on local hardware.

Failure 1: Resource Contention (ReadTimeoutError) *Error:* urllib3.exceptions.ReadTimeoutError

UnixHTTPConnectionPool... Read timed out.

Context: The OpenHands agent initialization requires constant communication between the Docker sandbox and the Host OS.

Failure: Local hardware resource contention (LLM inference + Docker runtime) caused system latency to exceed the 60-second connection timeout, triggering a crash during loop initialization.

Failure 2: State File Corruption (JSONDecodeError) *Error:* json.decoder.JSONDecodeError

Expecting value: line 1 column 1

Context: System timeouts forced the OpenHands server to terminate while writing conversation history.

Failure: This resulted in corrupted state JSON files. Reloading the system led to an infinite "Internal Server Error" loop, necessitating manual cleanup: `rm -rf ~/.openhands/state`.

Failure 3: Tool-Schema Mismatch (The security_risk Barrier) *Error:* Missing required parameters for function 'execute_bash': {'security_risk'}

Context: OpenHands v0.59 enforced a mandatory `security_risk` safety flag for all tool calls.

Failure: The local 8B model logically determined the correct Bash command but failed to inject the mandatory `"security_risk": "LOW"` JSON parameter into the background tool call.

Significance: This demonstrates that local models, while capable of high-level reasoning, currently lack the strict API-schema adherence required by modern sandbox environments.

4 Researcher Conclusion

This week's experiments confirm that local autonomous agent systems are constrained by two primary factors:

1. **Hardware Overhead:** The cumulative RAM requirement for local LLM inference + Sandbox Runtime exceeds standard consumer limits, leading to frequent "ReadTimeout" failures.
2. **Structural Brittleness:** Small-parameter models (under 10B) struggle to maintain strict adherence to evolving tool-schema requirements, rendering them unreliable for autonomous "QA/Tester" roles where tool execution is frequent.